Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **AI-assisted SDLC** workflows using Python, JavaScript, and cloud platforms, with strengths in **agent orchestration**, requirements translation, **automated testing**, CI/CD, and human-in-the-loop quality validation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Database Systems, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Restructured fault-tolerant **agent orchestration** with FastAPI workers, explicit job states, and failure isolation, translating complex processing requirements into reliable workflows that handled 1K+ AI jobs at triple throughput.
- Built recurring data workflows with validation, reconciliation, error handling, and CI/CD-backed releases, sustaining 99%+ **release reliability** across 5K+ monthly jobs while supporting 500+ users and documented operational requirements.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Developed structured **prompt workflows** connecting rubric generation, OCR extraction, grading, and feedback across GPT-4o, Claude, and Qwen-VL, delivering 90%+ grading consistency for 10K+ daily production requests at scale.
- Implemented concurrent batch processing, provider fallbacks, output validation, and CI/CD-driven releases, reducing feedback turnaround 60% while supporting **quality assurance** for 150+ instructors across rapidly changing product requirements.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Designed integration modules with object-oriented patterns and reviewed implementation choices for maintainability, improving response times 30%, reducing technical debt 20%, and reinforcing **engineering standards** across a microservice platform.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, and Jenkins across three operating systems, supporting Agile sprint delivery and release validation with Microsoft and GitHub stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built eight modular services for authentication, resources, billing, leads, permissions, and notifications, translating business workflows into maintainable APIs while synchronizing 2M+ records with 33% fewer **integration conflicts**.
- Participated in **code reviews** and iterative rollout while Jenkins CI/CD, Docker staging, and automated tests reached 95%+ build validation and reduced deployment time 50% across production release cycles.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Translated **client requirements** into an end-to-end architecture with 12+ integration points, mapping bookings, pricing, coupons, payments, and status workflows before presenting technical solutions for early stakeholder alignment.
- Built Flask APIs and real-time Firebase integrations for 50K+ booking records, validating payment and webhook workflows while maintaining sub-200ms responses and delivering **technical documentation** for cross-platform implementation.

Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML5, CSS3
AI & SDLC: Claude, GPT-4o, LLM APIs, Prompt Engineering, Context Design, Agent Orchestration, Requirements Analysis, Task Decomposition, Code Review
Quality: Unit Testing, Integration Testing, Release Validation, Output Evaluation, Documentation, Static Analysis, Monitoring
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, SQL Server, Azure SQL, MongoDB
Cloud & Delivery: AWS, Microsoft Azure, GCP, Docker, Jenkins, GitHub Actions, CI/CD, Git, Agile/Scrum, Sprint Execution, Stakeholder Communication

Projects

Agentic SDLC Review Console | *Python, Claude API, FastAPI, PyTest, GitHub Actions, Docker*
- Designed an **AI-assisted review workflow** that converts requirements into structured tasks, evaluates generated code, tests, and documentation against acceptance criteria, and routes incomplete artifacts through human approval gates.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*
- Built a LLaMA-2 **retrieval workflow** with transformer embeddings and FAISS over 50K+ documents, reliably providing semantic search and summarization for 200+ users while sustaining 95% query accuracy.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven, Node.js*
- Extended an AI-assisted Java and TypeScript workspace supporting 10+ databases, applying **structured validation** to schema-aware generated queries and Docker-based environments while achieving 82% accuracy against relational databases.